

2. The table shows some physical properties of Group 6 elements.

| Element   | Melting point (°C) | Boiling point (°C) | Density (g/cm <sup>3</sup> ) | Electrical conductor |
|-----------|--------------------|--------------------|------------------------------|----------------------|
| oxygen    | −219               | −183               | 0.0014                       | no                   |
| sulfur    | 115                | 445                | 2.0                          | no                   |
| selenium  | 221                | 685                | 4.8                          | semi-conductor       |
| tellurium | 450                | 988                | 6.2                          | semi-conductor       |

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

.....

(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

.....

.....

(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

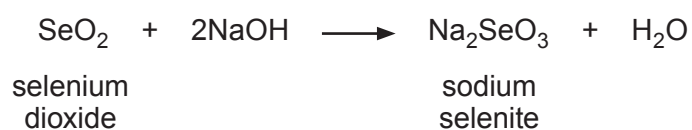
.....

.....

.....



(b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



(i) Calculate the relative formula mass ( $M_r$ ) of sodium selenite,  $\text{Na}_2\text{SeO}_3$ . [1]

$$A_r(\text{Na}) = 23 \qquad A_r(\text{Se}) = 79 \qquad A_r(\text{O}) = 16$$

$M_r = \dots\dots\dots$

(ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

Percentage =  $\dots\dots\dots$  %

3430UB01  
05

7

